

Zhenchen (Antonio) Dong

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Ph.D. Student, The Hong Kong Polytechnic University

Research Interests

Dynamic robotic manipulation with policies running on onboard edge chips; generative action policies; learning from large-scale robot data. Currently exploring space robotics with Prof. Hongyu Yu.

Education

The Hong Kong Polytechnic University Ph.D. in Mechanical Engineering; Advisor: Prof. Hongyu Yu	Jan 2026 – Present
The Hong Kong University of Science and Technology Doctoral studies in Mechanical and Aerospace Engineering	Aug 2025 – Jan 2026
Harbin Institute of Technology B.Eng. in Mechatronics GPA 90.1/100, Rank 1/28 Thesis: <i>Sim-to-Real Bimanual Robot Grasping Based on Imitation Learning</i>	Sep 2021 – Jun 2025

Publications & Patents

Zhenchen Dong*, Qingran Wu*, Jinna Fu, Jiaming Wu, Fulin Chen, Hongyu Yu, Yide Liu. *Shared Execution-Clock Drifting Policy for Dynamic Precision Manipulation*. Preprint, 2026.

Zhenchen Dong*, Fulin Chen*, Jinna Fu, Jiaming Wu, Qingran Wu, Shengyuan Yu, Hongyu Yu, Yide Liu. *HybridFlow: A 2-NFE Generative Policy for Real-Time Robotic Manipulation*. Preprint; arXiv:2602.13718, 2026.

*Equal contribution. Two invention patents, one utility model patent, and one design patent.

Industry Experience

Anker Innovations
Dexterous Manipulation Algorithm Intern

- Developed a VLA action expert module and generative policies for few-step and one-step robotic action generation

Galaxea AI
Embodied Intelligence Algorithm Intern

- Built a robotic simulation and evaluation platform; implemented Whole Body Control and object localization, grasping, and transfer with Relaxed IK and MpLib; designed LIBERO-based evaluation tasks and extended them to RoboVerse
- Calibrated and released URDF models for the R1 robot series, including asset reconstruction and simulation validation; the work was published in the official repository

Research Experience

State Key Lab of Robotics, HIT Mar 2024 – Nov 2024
Bipedal Robot Control & Research (Advisor: Prof. Haitao Yu)
Developed and researched bipedal robots in MuJoCo; focused on RL-based locomotion training and real-robot deployment

State Key Lab of Space Mechanisms, HIT Apr 2023 – Nov 2023
Zhurong Mars Rover Structural Verification (Advisor: Prof. Haibo Gao)
Analyzed inchworm locomotion and applied it to six-wheeled rover active suspension design; simulated terrain escape on slopes, craters, and regolith; performed mechanics analysis for each suspension configuration

Ruhr University Bochum Jul 2023 – Aug 2023
Solid Mechanics Summer Course (Advisor: Prof. A. Hartmaier)
Mechanical analysis of high-power high-density BLDC motor rotor materials

Skills

Engineering: PyTorch · Hardware Development · Git · Docker · ROS/ROS2 · Linux **Programming:** Python · MATLAB
Languages: English (fluent), Mandarin (native)

Beyond Research

Former starting player in the Floorball Asia Cup; skilled in mahjong and card games.

Selected Honors & Awards

Heilongjiang Provincial Merit Student	
National Scholarship; BYD Scholarship; UNIQLO Scholarship	
Grand Prize (National) , China Engineering Practice & Innovation Competition	2023
Grand Prize (National) , 25th China Robot & AI Competition	2023
1st Place , 9th TRCC National Robot Creative Design Competition	2022
National Project , Undergraduate Innovation Training — Underwater Pipeline Inspection Robot	2022